

HARSH KUMAR SINGH

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PROFESSIONAL SUMMARY

Backend Engineer with 3+ years of part-time, hands-on experience designing and building scalable, real-time backend systems in Python and FastAPI. Skilled in secure authentication architecture (JWT, OAuth), relational database design (PostgreSQL, MySQL), and containerized cloud deployment (Docker, AWS EC2). Completed an LLM evaluation internship at EtharaAI, with hands-on exposure to AI response evaluation and quality pipelines. Comfortable operating in Linux-based development environments with strong systems-level fluency.

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, Kotlin, JavaScript
- **Backend & Real-Time Frameworks:** FastAPI, Express.js, Next.js, REST APIs, WebRTC, Socket.IO, WebSockets
- **Frontend:** HTML, CSS, Tailwind CSS, JavaScript, Next.js
- **Databases:** PostgreSQL, MySQL, Firebase
- **Authentication & Security:** JWT, OAuth, Firebase Authentication, Zero-Trust API Security
- **Cloud & DevOps:** Docker, AWS EC2, Vercel, Render, Git, GitHub, CI/CD Deployment Workflows
- **Operating Systems:** Arch Linux (i3, KDE Plasma), Ubuntu 24.04 LTS, Windows

PROFESSIONAL EXPERIENCE

Assistant Store Manager | Blinkit (Eternal), Delhi

Aug 2026 - Aug 2026

- Managed day-to-day dark-store operations, coordinating order fulfillment and dispatch to maintain on-time delivery standards.
- Supervised store staff and delivery partners, maintaining productivity and service-quality standards during peak demand.
- Monitored inventory levels and replenishment cycles to minimize stock-outs and operational discrepancies.

LLM Post Trainer (Internship) | EtharaAI

Feb 2026 - Jul 2026

- Evaluated and improved large language model responses for real-world production scenarios across multiple datasets.
- Built structured evaluation and scoring workflows to guide prompt and response-quality improvements.
- Scraped and curated training data from web sources and GitHub pull requests to support model fine-tuning.
- Collaborated with the AI team on prompt analysis and quality-improvement pipelines.

Backend Engineer (Part-Time) | Kridavista

Dec 2022 - Jul 2026

- Developed scalable backend services for a real-time communication and collaboration platform using Python, FastAPI, and WebSocket-based architecture, supporting concurrent user sessions.
- Designed secure authentication and session-management systems using JWT and OAuth-based workflows to protect user accounts and data.
- Engineered low-latency media and messaging pipelines supporting concurrent user communication and streaming operations.
- Managed cloud deployments and containerized services across Docker, AWS EC2, Vercel, and Render to ensure reliable, always-on service.
- Optimized PostgreSQL database structures and backend APIs for improved scalability, response time, and resource efficiency.

Full Stack Developer (Part-Time) | SaiMadad

Aug 2024 - Oct 2025

- Built a scalable anonymous support and collaboration platform using Next.js, FastAPI, PostgreSQL, and secure authentication systems.
- Developed responsive frontend interfaces with optimized state management and API integrations.
- Implemented backend moderation pipelines and AI-assisted filtering systems for real-time chat safety.
- Designed RESTful APIs, database schemas, and secure user-access workflows for scalable platform operations.
- Integrated Firebase push notifications and JWT-based authorization mechanisms for protected services.

PROJECTS

Kridavista: Real-Time Communication Platform | WebRTC, Socket.IO, PostgreSQL, AWS EC2

- Architected a full-stack real-time communication platform end-to-end, from responsive interfaces to a scalable backend structure.
- Engineered low-latency bidirectional data streams using WebRTC and Socket.IO, optimizing server load and continuous media delivery for concurrent user sessions.
- Designed PostgreSQL schemas to manage relational user data, chat logs, and live-streaming metadata at scale.

SaiMadad: Anonymous Support Platform | PostgreSQL, Firebase Auth, JWT, AI Integration

- Led development of an open-source, anonymous support platform, integrating a responsive frontend with robust backend services.

- Implemented zero-trust authentication pipelines using Firebase and JWT to secure API endpoints and protect user data.
- Integrated AI-driven moderation to autonomously parse, evaluate, and filter real-time chat data streams for community safety.

HexaWave: Serverless, End-to-End Encrypted Android Messaging and Calling Platform | *Kotlin, Jetpack Compose, WebRTC, BouncyCastle*

- Architected a privacy-first Android app for encrypted messaging, file transfer, and voice calling directly between nearby devices over local Wi-Fi, with no backend server, cloud database, or third-party relay of any kind.
- Designed a complete cryptographic identity system, including Ed25519 signing, X25519 key agreement, Android Keystore-backed StrongBox private keys, and HKDF-derived AES-256-GCM/ChaCha20-Poly1305 session encryption, alongside five custom binary wire protocols with replay and tamper protection.
- Integrated local-network WebRTC voice calling with signaling carried over the app's own encrypted trust channel, and wrote over 100 JUnit tests covering cryptographic correctness, protocol encoding, and tamper and replay rejection.

SwiftDrop: Cross-Platform Peer-to-Peer File Transfer System | *Rust, Kotlin, Jetpack Compose, Noise Protocol, UniFFI*

- Built a private, serverless file-transfer system spanning a Windows and Linux CLI and an Android app, letting devices discover, pair, and exchange files directly over a local network with no cloud service or account required.
- Implemented end-to-end encrypted device pairing using the Noise protocol with human-verified short authentication codes, and a chunked, checksum-verified transfer engine with crash-safe resume.
- Bound a shared Rust transfer engine into a native Android app via UniFFI, including Keystore-backed key storage and QR-code pairing, and covered the system with over 70 automated tests, including real-socket integration tests.

Linux Music Player & Downloader | *Python, REST APIs, Linux CLI, Data Extraction*

- Built a lightweight, Linux-native multimedia data-extraction tool in Python to automate retrieval and processing of audio files without a graphical interface.
- Integrated the YouTube API to query, fetch, and transform streaming data into downloadable local formats.
- Optimized the application for minimal resource consumption using native command-line tooling and manual system configuration.

AlgoVerse: CPU Scheduling Simulator | *Algorithm Simulation, Data Visualization*

- Built an interactive CPU scheduling simulation and learning platform visualizing FCFS, SJF, Priority, and Round Robin algorithms.
- Implemented draggable process simulations, dynamic Gantt charts, and performance-metric displays to make scheduling algorithms interactive and easier to learn.

Additional projects available on GitHub: github.com/harshuopjs

EDUCATION

Guru Gobind Singh Indraprastha University, Delhi, India B.Tech in Computer Science Engineering	2023 - 2027 (Expected)
Govt. Sarvodaya Bal Vidyalaya, Delhi, India Senior Secondary School (Class 11-12)	Completed 2023
Joseph and Mary Public School, Delhi, India Secondary School (Class 10)	Completed 2021

CERTIFICATIONS

- Google Cybersecurity Professional Certificate, Coursera
- Web Development Masterclass, Udemy
- Java Programming Certificate
- Mid-Level JavaScript Developer, Certificates.dev

ACHIEVEMENTS

- Felicited and Awarded at BharatPreneurs 2026
- Selected for Pre-Incubation at DTU-IIF
- 1st Place, Internal Hackathon (Smart India Hackathon 2025), JIMS EMTC
- 9th Place, Internal Hackathon (Smart India Hackathon 2024), JIMS EMTC
- Runner-up, VCT Hashtag Competition (Team WARDEN)
- IIC Activity Award, National Startup Day